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On the uniqueness of the shape of a penetrable, anisotropic obstacle

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Abstract

We consider the scattering of time-harmonic waves in an isotropic, homogeneous medium which contains a penetrable, anisotropic, inhomogeneous obstacle. After reviewing existence, uniqueness and regularity of a (variational) solution we prove that a knowledge of the scattered near fields for all incident plane waves at a fixed wave number uniquely determines the obstacle provided the matrix characterizing the anisotropic medium satisfies a suitable condition. To this end we examine the corresponding interior transmission problem and prove that it is a compact perturbation of an interior transmission problem which is well-posed. © 2000 Elsevier Science B.V. All rights reserved.

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1. Introduction

In this paper we are interested in the shape of a penetrable, anisotropic, inhomogeneous obstacle, which is surrounded by a homogeneous, isotropic medium, assuming that we know near field data corresponding to plane incident waves for all incident directions at a fixed wave number.

The usual mathematical model describing the scattering of time-harmonic waves by a penetrable obstacle is a transmission problem. The transmission problem that we have in mind is motivated by the transmission problem which models the scattering of time-harmonic, electromagnetic, H -polarized waves by an infinitely long cylinder filled with an orthotropic dielectric. For this scattering problem, in [3,14,1] the authors derive the two-dimensional analog to the problem we shall formulate in the next section.

Besides the formulation of the direct problem the next section also contains a review of the results for the direct problem which we shall need for the inverse problem, i.e., the well-posedness of the direct problem and a regularity result for solutions of the direct problem.

The final section is devoted to the uniqueness proof for the inverse problem. The main idea for our uniqueness proof originates from Isakov's paper [10], where he identifies a region having a different conductivity from the surrounding medium. Assuming two different regions produce the same voltage and current measurements on the surface of the exterior medium, he derives a contradiction by using a sequence of functions having a singularity which moves closer and closer to the boundary of one region while staying away from the other region. In [11] he also employs this idea to prove that the shape of a penetrable, homogeneous, isotropic obstacle is uniquely determined by its far fields for all incident plane waves. He uses weak solutions and his proofs are technically involved. In [13] Kirsch and Kress simplified his analysis using classical solutions and integral equation methods. However, they still need a tedious analysis of boundary integral operators and boundary integral equations. Boundary integral equations are a suitable tool to analyze partial differential equations with constant coefficients, but they are not well tailored for partial differential equations having inhomogeneous coefficients in front of the highest derivatives. Consequently, in [3], where the idea from [13] is worked out for a homogeneous, anisotropic obstacle in two dimensions, the authors restrict their analysis to the constant coefficient case. Piana has generalized their result and has treated a two-dimensional, inhomogeneous, anisotropic obstacle in [14]. He uses the Lippmann–Schwinger equation to construct the fundamental solution for the anisotropic medium and then employs the boundary integral equation techniques from [3] with this fundamental solution. Again, due to the use of boundary integral equation methods, he must assume that the matrix describing the anisotropic medium is a constant matrix along the boundary.

A well-known approach for partial differential equations with inhomogeneous coefficients is the variational method, which is not only used for a theoretical treatment but is also well established as the finite-element method for the numerical treatment. Therefore, as in Isakov's paper [10], in this paper we try again to bring together the variational approach and his idea. However, we restructure the proof in such a way that it can be carried through provided the direct problem is well-posed, there is a regularity theorem for the direct problem, and the interior transmission problem related to the direct problem is a compact perturbation of a well-posed problem. This structure of the proof hopefully turns out to be useful in other inhomogeneous medium problems, too. We also want to emphasize that our analysis is independent of the dimension, although we have restricted ourselves to the three-dimensional case.

Of course, we avoid boundary integral equations. To be honest, we still obtain an equation on the boundary during the analysis of the interior transmission problem. However, with the help of assumption (4) we can discuss this equation without ever using boundary integral operators. The third section is devoted to the interior transmission problem. We will formulate a modified interior transmission problem, prove its well-posedness, and then, by a compact perturbation argument, obtain the needed result about the interior transmission problem related to the direct transmission problem. Let us finally remark that interior transmission problems naturally appear in inverse inhomogeneous medium problems. For example they determine whether the range of the far-field operator is dense (see [2, Theorem 8.9] and references given there) and in [4] the authors employ them for the analysis of their reconstruction procedure of D from far field data. Therefore, our analysis is useful, too, if one needs similar results for the transmission problem with an anisotropic obstacle.

2. The direct problem

Let $D \subset \mathbb{R}^3$ be a nonempty, open, and bounded set having a C^2 -smooth boundary ∂D . The unit normal vector to ∂D , which is directed into the exterior of D , is denoted by v . Moreover, we assume that the exterior domain $D_e := \mathbb{R}^3 \setminus \bar{D}$ is connected. On $\bar{D}$ we have a matrix-valued function $A: \bar{D} \rightarrow \mathbb{C}^{3 \times 3}$, $A = (a_{jk})_{j,k=1,\dots,3}$, with continuously differentiable functions $a_{jk} \in C^1(\bar{D})$. By $\Re(A)$ we mean the matrix-valued function having as entries the real parts $\Re(a_{jk})$, and similarly we define $\Im(A)$. We always suppose that $\Re(A(x))$ and $\Im(A(x))$, $x \in \bar{D}$, are symmetric matrices which satisfy $\xi^* \cdot \Im(A(x)) \xi \leq 0$ and $\xi^* \cdot \Re(A(x)) \xi \geq \alpha |\xi|^2$ with a positive constant $\alpha > 0$ for all $\xi \in \mathbb{C}^3$ and all $x \in \bar{D}$.

For a function $u \in C^1(\bar{D})$ we define the conormal derivative

$$\frac{\partial u}{\partial v_A}(x) := v(x) \cdot A(x) \nabla u(x), \quad x \in \partial D.$$

By $H^k(G)$ we mean the usual Sobolev space of square integrable functions in an open set G possessing square integrable weak derivatives up to order k with its standard norm. For a C^2 -smooth boundary the trace theorem ensures that functions from $H^1(G)$ possess boundary values in the Sobolev space $H^{1/2}(\partial G)$. $H^{1/2}(\partial G)$ can be equipped with any scalar product such that the trace operator is bounded and has a bounded right-inverse.

With $H_0^k(G)$ being the completion of $C_0^\infty(G)$ in $H^k(G)$ we have

$$H_0^1(G) = \{u \in H^1(G): u|_{\partial G} = 0\}.$$

Now, we can generalize the definition of the conormal derivative to any function $u \in H^1(D)$ which satisfies $-\operatorname{div}(A \nabla u) = h \in L^2(D)$ in the weak sense (see [5, Lemma 3.2]), i.e.

$$\int_D \nabla \varphi \cdot A \nabla u \, dx = \int_D h \varphi \, dx$$

for all $\varphi \in C_0^\infty(D)$. Namely, for a function $f \in H^{1/2}(\partial D)$ we set

$$\int_{\partial D} \frac{\partial u}{\partial v_A} f \, ds := \int_D \nabla \psi \cdot A \nabla u \, dx - \int_D h \psi \, dx$$

where $\psi \in H^1(D)$ is an arbitrary function with $\psi|_{\partial D} = f$. The right-hand side is independent of ψ and can be bounded by the H^1 -norm of ψ . Hence, it is a bounded linear functional on $H^{1/2}(\partial D)$ which we denote by $\partial u / \partial v_A$. Note that the integral on the left has to be understood in the sense of the dual pairing between $H^{1/2}(\partial D)$ and its dual space $H^{-1/2}(\partial D)$ with pivot space $L^2(\partial D)$. Similarly, we can define $\partial v / \partial v$ on ∂D for a function $v \in H_{\text{loc}}^1(D_e)$ with $\Delta v \in L_{\text{loc}}^2(D_e)$ in the weak sense.

We are now in a position to formulate the direct scattering problem (TP): given the set D , the wave number $\kappa > 0$, a continuous function $n \in C(\bar{D})$ with $\Im(n) \geq 0$, a matrix-valued function $A: \bar{D} \rightarrow \mathbb{C}^{3 \times 3}$ as above, and boundary data $f \in H^{1/2}(\partial D)$, $g \in H^{-1/2}(\partial D)$, find functions $u \in H^1(D)$ and $w \in H_{\text{loc}}^1(D_e)$ satisfying

$$\Delta w + \kappa^2 w = 0 \text{ in } D_e, \quad \operatorname{div}(A \nabla u) + \kappa^2 n u = 0 \text{ in } D \text{ (in the weak sense),}$$

the radiation condition $|x \cdot \nabla w(x) - i\kappa|x|w(x)| \rightarrow 0$, $|x| \rightarrow \infty$, and the boundary conditions

$$u|_{\partial D} - w|_{\partial D} = f, \quad \frac{\partial u}{\partial \nu_A} - \frac{\partial w}{\partial \nu} = g \quad \text{on } \partial D.$$

Since by standard regularity results a weak solution to the Helmholtz equation in D_e is actually in $C^2(D_e)$ (see [7, Corollary 8.11]), the radiation condition makes sense.

As already mentioned in the introduction the corresponding problem in $\mathbb{R}^2$ can be used as a model to describe the scattering of TE-polarized electromagnetic waves by an orthotropic, infinitely long, dielectric cylinder. In [15] the problem is analyzed for the two-dimensional case with the help of a singular integral equation, obtained from the Lippmann–Schwinger equation, assuming additionally that extending A by the identity matrix into D_e yields a continuously differentiable matrix-valued function in all of $\mathbb{R}^2$. In [14] Piana uses a boundary integral equation with the additional assumption that A coincides with a constant matrix on the boundary. Our variational formulation can dispense with these restrictions.

In order to arrive at a variational formulation over a bounded domain, we introduce a large, open ball B containing $\bar{D}$. Furthermore, we choose a function $v_f \in H^1(B \setminus \bar{D})$ with the boundary values $f \in H^{1/2}(\partial D)$ on ∂D by solving the Dirichlet problem

$$\Delta v_f - v_f = 0 \quad \text{in } B \setminus \bar{D} \text{ (in the weak sense),} \quad v_f|_{\partial D} = f, \quad v_f|_{\partial B} = 0. \quad (1)$$

And finally, we introduce the Dirichlet-to-Neumann map

$$A: H^{1/2}(\partial B) \rightarrow H^{-1/2}(\partial B)$$

which maps $h \in H^{1/2}(\partial B)$ to $\partial \tilde{w}/\partial \nu$ where $\tilde{w}$ solves the exterior Dirichlet problem for the Helmholtz equation $\Delta \tilde{w} + \kappa^2 \tilde{w} = 0$ in $\mathbb{R}^3 \setminus \bar{B}$ with Dirichlet boundary data $\tilde{w}|_{\partial B} = h$ (see [12] or [2, p. 116] for details).

Then, the variational formulation (VTP) of the transmission problem (TP) reads: given D , A , n , κ , f , and g , find $u \in H^1(B)$ satisfying

$$\begin{aligned} \int_D (\nabla \bar{\varphi} \cdot A \nabla u - \kappa^2 n \bar{\varphi} u) \, dx + \int_{B \setminus \bar{D}} (\nabla \bar{\varphi} \cdot \nabla u - \kappa^2 \bar{\varphi} u) \, dx - \int_{\partial B} \bar{\varphi} A u \, ds \\ = \int_{\partial D} \bar{\varphi} g \, ds - \int_{\partial B} \bar{\varphi} A v_f \, ds + \int_{B \setminus \bar{D}} (\nabla \bar{\varphi} \cdot \nabla v_f - \kappa^2 \bar{\varphi} v_f) \, dx \end{aligned} \quad (2)$$

for all $\varphi \in H^1(B)$. Note that all surface integrals denote the duality pairing between $H^{1/2}$ and $H^{-1/2}$. The operator A replaces the radiation condition by a nonlocal boundary condition on the artificial boundary ∂B .

If u is a solution to (VTP), it is easily seen that $u|_D$ and $w := u|_{B \setminus \bar{D}} - v_f$ satisfy the differential equations, the boundary conditions from the original transmission problem (TP), and $\partial w/\partial \nu = A w$ on ∂B .

In order to define w in all of $\mathbb{R}^3 \setminus \bar{D}$ we use Green's second theorem and the radiation condition to see that

$$\int_{\partial B} \left((A h)(y) \Phi(x, y) - h(y) \frac{\Phi(x, y)}{\partial \nu(y)} \right) \, ds(y) = 0, \quad x \in B,$$

for all $h \in C^{1,\alpha}(\partial B)$, whence for all $h \in H^{1/2}(\partial B)$. Here, $\Phi(x, y)$ denotes the fundamental solution

$$\Phi(x, y) := \frac{e^{ik|x-y|}}{4\pi|x-y|}, \quad x, y \in \mathbb{R}^3, \quad x \neq y. \quad (3)$$

Consequently, the representation theorem (see [5, Lemma 3.4]) and $\partial w / \partial v = \Lambda w$ imply

$$\begin{aligned} w(x) &= \int_{\partial D} \left(w(y) \frac{\partial \Phi(x, y)}{\partial v(y)} - \frac{\partial w}{\partial v}(y) \Phi(x, y) \right) ds(y) \\ &\quad + \int_{\partial B} \left(\frac{\partial w}{\partial v}(y) \Phi(x, y) - w(y) \frac{\partial \Phi(x, y)}{\partial v(y)} \right) ds(y) \\ &= \int_{\partial D} \left(w(y) \frac{\partial \Phi(x, y)}{\partial v(y)} - \frac{\partial w}{\partial v}(y) \Phi(x, y) \right) ds(y), \quad x \in B \setminus \bar{D}, \end{aligned}$$

and w coincides with a radiating solution to the Helmholtz equation in the exterior of $\bar{D}$. Hence, a solution to (TP) can be derived from a solution to (VTP).

Our first theorem states the well-posedness of the problems (TP) and (VTP).

Theorem 1. *The problems (TP) and (VTP) possess a unique solution which depends continuously on the data f and g .*

Proof. Let us begin with the uniqueness of a solution. According to our previous remarks, a solution to the homogeneous problem (VTP) ($f = 0$, $g = 0$) can be extended to a solution $u \in H^1(D)$ and $w \in H_{\text{loc}}^1(D_e)$ of the homogeneous problem (TP). Green's theorem and the boundary conditions imply

$$\begin{aligned} \int_{\partial B} \bar{w} \frac{\partial w}{\partial v} ds &= \int_{\partial D} \bar{w} \frac{\partial w}{\partial v} ds + \int_{B \setminus \bar{D}} (|\nabla w|^2 - \kappa^2 |w|^2) dx \\ &= \int_D (\nabla \bar{u} \cdot A \nabla u - \kappa^2 n |u|^2) dx + \int_{B \setminus \bar{D}} (|\nabla w|^2 - \kappa^2 |w|^2) dx, \end{aligned}$$

whence

$$\Im \left\{ \int_{\partial B} \bar{w} \frac{\partial w}{\partial v} ds \right\} \leq 0.$$

With the help of Theorem 2.12 in [2] we can conclude $w = 0$ in D_e , and then $u|_{\partial D} = 0$, and $\partial u / \partial v_A = 0$ on ∂D by the boundary conditions. In order to show $u = 0$ we employ a unique continuation principle. We extend $\Re(A)$ to a real, symmetric, positive definite, and continuously differentiable matrix-valued function in $\bar{B}$ and $\Im(A)$ to a real, symmetric, continuously differentiable matrix-valued function which is compactly supported in B . We also choose a continuously differentiable extension of n into $\bar{B}$ and define $u(x) = 0$ for $x \in B \setminus D$. Then $u \in H^1(B)$ is a weak solution to $\operatorname{div}(A \nabla u) + \kappa^2 n u = 0$ in B and therefore it is in $H^2(B)$ by the usual regularity theorems (see [6, Theorem 15.1]). Now, we can apply the unique continuation principle from Theorem 17.2.6 in [9] and obtain $u = 0$.

In order to prove existence of a solution to (VTP) we also introduce the Dirichlet-to-Neumann map Λ_0 for the Laplace equation

$$\Lambda_0 : H^{1/2}(\partial B) \rightarrow H^{-1/2}(\partial B)$$

which maps $h \in H^{1/2}(\partial B)$ to $\partial \tilde{w}/\partial \nu$ where $\tilde{w}$ solves the exterior Dirichlet problem for the Laplace equation in $\mathbb{R}^3 \setminus \bar{B}$ with Dirichlet boundary data $\tilde{w}|_{\partial B} = h$. It is known that A_0 satisfies

$$-\int_{\partial B} \bar{h} A_0 h \, ds \geq 0$$

for all $h \in H^{1/2}(\partial B)$ (the integral denotes again the duality pairing), and that the difference $A - A_0$ is a compact operator from $H^{1/2}(\partial B)$ to $H^{-1/2}(\partial B)$ (see [2, p. 116] or [12]).

Defining the sesquilinear forms

$$a(\varphi, \psi) := \int_D \nabla \bar{\varphi} \cdot A \nabla \psi \, dx + \int_{B \setminus \bar{D}} \nabla \bar{\varphi} \cdot \nabla \psi \, dx + \int_B \bar{\varphi} \psi \, dx - \int_{\partial B} \bar{\varphi} A_0 \psi \, ds$$

and

$$a_1(\varphi, \psi) := - \int_D (\kappa^2 n + 1) \bar{\varphi} \psi \, dx - (\kappa^2 + 1) \int_{B \setminus \bar{D}} \bar{\varphi} \psi \, dx - \int_{\partial B} \bar{\varphi} (A - A_0) \psi \, ds$$

for $\varphi, \psi \in H^1(B)$ Eq. (2) can be rewritten

$$a(\varphi, u) + a_1(\varphi, u) = \int_{\partial D} \bar{\varphi} g \, ds - \int_{\partial B} \bar{\varphi} A v_f \, ds + \int_{B \setminus \bar{D}} (\nabla \bar{\varphi} \cdot \nabla v_f - \kappa^2 \bar{\varphi} v_f) \, dx.$$

The Riesz representation theorem ensures the existence of a function $r \in H^1(B)$ such that the right-hand side of the above equation coincides with $(\varphi, r)_{H^1}$ for all $\varphi \in H^1(B)$ where $(\cdot, \cdot)_{H^1}$ denotes the usual scalar product in $H^1(B)$. As a consequence of the Lax–Milgram theorem (see [6, Theorem 14.1]) and the Riesz representation theorem there is a bounded, linear operator $T: H^1(B) \rightarrow H^1(B)$ having a bounded inverse such that $(\varphi, T\psi)_{H^1} = a(\varphi, \psi)$ holds true for all $\varphi, \psi \in H^1(B)$. Similarly, there is a bounded, linear operator T_1 in $H^1(B)$ with $(\varphi, T_1\psi)_{H^1} = a_1(\varphi, \psi)$. Due to the compactness of $A - A_0$ and of the embedding of $H^1(B)$ into $L^2(B)$ we can conclude that weakly convergent sequences $\varphi_k \rightharpoonup \varphi$, $\psi_k \rightharpoonup \psi$, $k \rightarrow \infty$, satisfy $(\varphi_k, T_1\psi_k)_{H^1} \rightarrow (\varphi, T_1\psi)_{H^1}$, $k \rightarrow \infty$. Consequently, for any weakly convergent sequence $(\psi_k)_k$, the sequence $\varphi_k := T_1\psi_k$ is weakly convergent and the norms $\|T_1\psi_k\|_{H^1}^2$ converge to $\|T_1\psi\|_{H^1}^2$, i.e., $(T_1\psi_k)_k$ strongly converges to $T_1\psi$. Therefore, the operator T_1 is compact. Hence, Eq. (2) is equivalent to the equation $Tu + T_1u = r$ in $H^1(B)$ which can be treated by the Riesz–Fredholm alternative. Since we have already shown that the homogeneous equation possesses only the trivial solution, we know that the inhomogeneous equation has a unique solution which depends continuously on the right-hand side, i.e., on f and g . This completes the proof for the problem (VTP). Noting that a solution to (VTP) can be extended to a solution of (TP) ends the proof of the theorem. $\square$

Our next theorem deals with the local boundary regularity of a solution to (TP). By $B_\varepsilon(z)$ we mean an open ball centered at $z \in \mathbb{R}^3$ with radius ε .

Theorem 2. Assume $z \in \partial D$, B is a large, open ball containing $\bar{D}$, and $v \in H^1(D)$ satisfies $\Delta v \in L^2(D)$ in the weak sense. Define $f := v|_{\partial D}$, $g := \partial v / \partial \nu$.

(a) Moreover, for some $\varepsilon > 0$ assume v is also defined in $B_{2\varepsilon}(z)$ and $v|_{B_{2\varepsilon}(z)} \in H^2(B_{2\varepsilon}(z))$. If ε is sufficiently small, then the solution w to (TP) satisfies $w|_{D_\varepsilon \cap B_\varepsilon(z)} \in H^2(D_\varepsilon \cap B_\varepsilon(z))$ and there is a positive constant C such that

$$\|w\|_{H^2(D_\varepsilon \cap B_\varepsilon(z))} \leq C \{ \|v\|_{H^2(B_{2\varepsilon}(z))} + \|v\|_{H^1(D)} \}.$$

- (b) If for some $\varepsilon > 0$ the incident wave v is also defined in $B \setminus B_\varepsilon(z)$ and $v|_{B \setminus B_\varepsilon(z)} \in H^2(B \setminus B_\varepsilon(z))$, then the solution w to (TP) satisfies $w|_{(B \cap D_e) \setminus B_{2\varepsilon}(z)} \in H^2((B \cap D_e) \setminus B_{2\varepsilon}(z))$ and there is a positive constant C such that

$$\|w\|_{H^2((D_e \cap B) \setminus B_{2\varepsilon}(z))} \leq C\{\|v\|_{H^2(B \setminus B_\varepsilon(z))} + \|v\|_{H^1(D)}\}.$$

Proof. For the proof of part (a), without loss of generality we suppose that there is a C^2 -smooth diffeomorphism $\Psi: B_{2\varepsilon}(z) \rightarrow G \subset \mathbb{R}^3$ which straightens the boundary, i.e., $G_+ := \Psi(D_e \cap B_{2\varepsilon}(z)) = \{y \in G: y_3 > 0\}$, $G_- := \Psi(D \cap B_{2\varepsilon}(z)) = \{y \in G: y_3 < 0\}$, and $\Psi(\partial D \cap B_{2\varepsilon}(z)) = \{y \in G: y_3 = 0\}$. Here, we need ε sufficiently small in order to have such a diffeomorphism.

Defining $u(x) := v(x) + w(x)$ for $x \in D_e \cap B_{2\varepsilon}(z)$ we have $u \in H^1(B_{2\varepsilon}(z))$ and u satisfies

$$\int_{D \cap B_{2\varepsilon}(z)} (\nabla \varphi \cdot A \nabla u - \kappa^2 n \varphi u) dx + \int_{D_e \cap B_{2\varepsilon}(z)} (\nabla \varphi \cdot \nabla u - \kappa^2 \varphi u) dx = \int_{D_e \cap B_{2\varepsilon}(z)} \varphi h dx$$

for all $\varphi \in H_0^1(B_{2\varepsilon}(z))$, where $h := -\Delta v - \kappa^2 v \in L^2(D_e \cap B_{2\varepsilon}(z))$.

Substituting $x = \Psi^{-1}(y)$, $y \in G$, we obtain for $u' := u \circ \Psi^{-1} \in H^1(G)$:

$$\int_{G_-} (\nabla \varphi \cdot A'_- \nabla u' - \kappa^2 n'_- \varphi u') dx + \int_{G_+} (\nabla \varphi \cdot A'_+ \nabla u' - \kappa^2 n'_+ \varphi u') dx = \int_{G_+} \varphi h' dx$$

for all $\varphi \in H_0^1(G)$, where the matrix-valued functions A'_+ and A'_- still satisfy the analogous assumptions to A .

Next, we set $G' := \Psi(B_\varepsilon(z))$, and therefore $\bar{G}' \subset G$. If we copy the standard proof for the derivation of higher-order derivatives in the interior of G , i.e., we estimate the norms of difference quotients in y_1 - or y_2 -direction (see [7, Proof of Theorem 8.8] for details), we see that $\partial^2 u' / \partial y_j \partial y_k \in L^2(G')$ for $j, k = 1, 2, 3$, $j \neq 3$ or $k \neq 3$, hold true. Furthermore, the $L^2(G')$ -norms are bounded by a multiple of the $L^2(G)$ -norm of h' and the $H^1(G)$ -norm of u' . The existence of the derivative $\partial^2 u' / \partial y_3^2$ in $G' \cap \{y_3 > 0\}$ is then established with the help of the differential equation:

$$\sum_{j,k=1}^3 \partial_j (a'_{jk+} \partial_k u') = \operatorname{div}(A'_+ \nabla u') \in L^2(G')$$

can be solved for $\partial_3^2 u' \in L^2(G' \cap \{y_3 > 0\})$ because, due to the positive definiteness of $\Re(A'_+)$, $\Re(a'_{33+}(x)) \geq \alpha' > 0$ holds true for all $x \in G \cap \{y_3 > 0\}$. Hence, we have $u'|_{G' \cap \{y_3 > 0\}} \in H^2(G' \cap \{y_3 > 0\})$ and the estimate

$$\|u'\|_{G' \cap \{y_3 > 0\}} \|h'\|_{H^2(G' \cap \{y_3 > 0\})} \leq C(\|u'\|_{H^1(G)} + \|h\|_{L^2(G)}).$$

Finally, we transform back $(w + v)|_{D_e \cap B_\varepsilon(z)} = (u' \circ \Psi)|_{D_e \cap B_\varepsilon(z)}$ and obtain the desired regularity of w together with the norm estimate.

For assertion (b) we use part (a) at the boundary and the standard interior regularity result (see [7, Proof of Theorem 8.8]) away from the boundary. $\square$

3. The interior transmission problem

Before dealing with the inverse problem we examine the interior transmission problem (ITP) corresponding to the transmission problem of the previous section. The problem (ITP) reads:

given D , κ , n , and A as in the direct problem and given $f \in H^{1/2}(\partial D)$, $g \in H^{-1/2}(\partial D)$, find two functions u , $v \in H^1(D)$ satisfying

$$\operatorname{div}(A\nabla u) + \kappa^2 nu = 0, \quad \Delta v + \kappa^2 v = 0 \quad \text{in } D \text{ (in the weak sense),}$$

and the boundary conditions

$$u|_{\partial D} - v|_{\partial D} = f, \quad \frac{\partial u}{\partial v_A} - \frac{\partial v}{\partial v} = g \quad \text{on } \partial D.$$

Our goal is to show that, under suitable restrictions on A , a slightly modified interior transmission problem has a unique solution which depends continuously on the data f and g . In turn, this modified interior transmission problem allows to derive the necessary results for solutions of (ITP), which we want to employ during the analysis of the inverse problem in the last section.

From now on we assume that there is a constant $\gamma \in \mathbb{C}$, $|\gamma| > 1$, $\Re(\gamma) \geq 0$, $\Im(\gamma) \leq 0$, such that the inequality

$$\Re \left\{ \frac{1}{\gamma} \xi^* \cdot A(x) \xi \right\} \geq |\xi|^2 \quad (4)$$

holds true for all $x \in \bar{D}$ and all $\xi \in \mathbb{C}^3$. For example, if we know $\Re(A(x)) \geq \gamma_0$ with $\gamma_0 > 1$ in $\bar{D}$, we can choose $\gamma = \gamma_0$. Similarly, we choose $\gamma = -i\gamma_0$ for $\Im(A(x)) \leq -\gamma_0$ with $\gamma_0 > 1$. With the help of the constant γ we formulate the modified interior transmission problem (MITP):

given D , κ , n , and A as in the direct problem and given $f \in H^{1/2}(\partial D)$, $g \in H^{-1/2}(\partial D)$, $h_1, h_2 \in L^2(D)$, find two functions u , $v \in H^1(D)$ satisfying

$$\operatorname{div}(A\nabla u) - \gamma u = h_1, \quad \Delta v - v = h_2 \quad \text{in } D \text{ (in the weak sense),}$$

and the boundary conditions

$$u|_{\partial D} - v|_{\partial D} = f, \quad \frac{\partial u}{\partial v_A} - \frac{\partial v}{\partial v} = g \quad \text{on } \partial D.$$

Note that compared to (ITP) we have modified the differential equations. We also allow inhomogeneities h_1, h_2 to use compact perturbation arguments for (ITP) later.

In order to show the unique solvability of (MITP) we introduce the Dirichlet-to-Neumann maps $A_A: H^{1/2}(\partial D) \rightarrow H^{-1/2}(\partial D)$, $A_A: H^{1/2}(\partial D) \rightarrow H^{-1/2}(\partial D)$, with $A_A f = \partial v / \partial v$, where $v \in H^1(D)$ is the weak solution to $\Delta v - v = 0$ in D and $v|_{\partial D} = f$, and with $A_A f = \partial u / \partial v_A$, where $u \in H^1(D)$ is the weak solution to $\operatorname{div}(A\nabla u) - \gamma u = 0$ in D having boundary data $u|_{\partial D} = f$. Note that the assumptions on γ ensure the existence of u . Since the Neumann problem $\operatorname{div}(A\nabla u) - \gamma u = 0$ in D , $\partial u / \partial v_A = g \in H^{-1/2}(\partial D)$ also possesses a unique solution, the Neumann-to-Dirichlet map A_A^{-1} exists, too.

We base our analysis of the problem (MITP) on the fact that $A_A^{-1} A_A$ is a contraction. This is the assertion of the next lemma.

Lemma 3. *Suppose that Assumption (4) is satisfied. If $H^{1/2}(\partial D)$ is equipped with the norm*

$$\|f\|_{1/2} := \inf \{ \|v\|_{H^1(D)} : v|_{\partial D} = f \},$$

then $A_A^{-1} A_A: H^{1/2}(\partial D) \rightarrow H^{1/2}(\partial D)$ is a contraction. Especially, the operator $I - A_A^{-1} A_A$ has a bounded inverse with respect to any norm on $H^{1/2}(\partial D)$ for which the trace of $H^1(D)$ -functions is well defined, continuous, and has a continuous right inverse.

Proof. Let us first note that the norm $\|\cdot\|_{1/2}$ is equivalent to any norm $\|\cdot\|$ on $H^{1/2}(\partial D)$ for which the trace is well-defined, continuous, and has a continuous right inverse. If v denotes the weak solution to $\Delta v - v = 0$ in D with boundary values $v|_{\partial D} = f$, then $\|v\|_{H^1(D)} = \inf\{\|w\|_{H^1(D)} : w|_{\partial D} = f\}$, and we can conclude $\|f\|_{1/2} = \|v\|_{H^1(D)} \leq C_1 \|f\|$ by the continuity of the right inverse $H^{1/2}(\partial D) \rightarrow H^1(D)$. On the other hand the continuity of the trace mapping implies $\|f\| \leq C_2 \|v\|_{H^1(D)} = C_2 \|f\|_{1/2}$.

Next, we prove $\|\partial v / \partial \nu\|_{-1/2} = \|v|_{\partial D}\|_{1/2}$ for any solution $v \in H^1(D)$ to $\Delta v - v = 0$. Here, $\|\cdot\|_{-1/2}$ denotes the usual norm on the dual space of $(H^{1/2}(\partial D), \|\cdot\|_{1/2})$. Given $f \in H^{1/2}(\partial D)$ let $w \in H^1(D)$ be the solution to $\Delta w - w = 0$ in D with $w|_{\partial D} = f$. Green's theorem implies

$$\left| \int_{\partial D} \frac{\partial v}{\partial \nu} f \, ds \right| = \left| \int_D (\nabla v \cdot \nabla w + vw) \, dx \right| \leq \|v\|_{H^1(D)} \|w\|_{H^1(D)} = \|v|_{\partial D}\|_{1/2} \|f\|_{1/2},$$

and therefore $\|\partial v / \partial \nu\|_{-1/2} \leq \|v|_{\partial D}\|_{1/2}$. Inserting $f = v|_{\partial D}$ we see

$$\left\| \frac{\partial v}{\partial \nu} \right\|_{-1/2} = \|v|_{\partial D}\|_{1/2}.$$

Now, starting with an arbitrary $f \in H^{1/2}(\partial D)$ we set $v \in H^1(D)$ to be the solution to $\Delta v - v = 0$ in D with $v|_{\partial D} = f$. Moreover, we denote by $u \in H^1(D)$ the solution to $\operatorname{div}(A \nabla u) - \gamma u = 0$ in D with $\partial u / \partial \nu_A = \partial v / \partial \nu$ on ∂D . Hence, we have $u|_{\partial D} = A_A^{-1} A_A f$ and we can estimate

$$\begin{aligned} |\gamma| \|A_A^{-1} A_A f\|_{1/2}^2 &= |\gamma| \|u|_{\partial D}\|_{1/2}^2 \\ &\leq |\gamma| \|u\|_{H^1(D)}^2 \\ &\leq \Re \left\{ \int_D \frac{|\gamma|}{\gamma} (\nabla \bar{u} \cdot A \nabla u + \gamma |u|^2) \, dx \right\} \\ &\leq \left| \int_D (\nabla \bar{u} \cdot A \nabla u + \gamma |u|^2) \, dx \right| \\ &= \left| \int_{\partial D} \frac{\partial u}{\partial \nu_A} \bar{u} \, ds \right| \\ &\leq \left\| \frac{\partial v}{\partial \nu} \right\|_{-1/2} \|u|_{\partial D}\|_{1/2} \\ &= \|f\|_{1/2} \|A_A^{-1} A_A f\|_{1/2}, \end{aligned}$$

i.e., $\|A_A^{-1} A_A\|_{1/2} \leq 1/|\gamma| < 1$.

The second part is an immediate consequence of the Banach fixed point theorem. $\square$

Let us remark that the preceding lemma examines the boundary operator $A_A^{-1} A_A$ without explicitly computing boundary integral operators.

Now, we are in a position to prove the existence of a unique solution to (MITP).

Theorem 4. Suppose A satisfies assumption (4). Then, for any given $f \in H^{1/2}(\partial D)$, $g \in H^{-1/2}(\partial D)$, $h_1, h_2 \in L^2(D)$ the modified interior transmission problem (MITP) has a unique solution

$u, v \in H^1(D)$. Furthermore, with a suitable constant C the inequality

$$\|u\|_{H^1(D)} + \|v\|_{H^1(D)} \leq C\{\|f\|_{1/2} + \|g\|_{-1/2} + \|h_1\|_{L^2} + \|h_2\|_{L^2}\}$$

holds true.

Proof. If u and v denote the solution of the homogeneous problem, we have

$$v|_{\partial D} = u|_{\partial D} = A_A^{-1}(\partial u / \partial v_A) = A_A^{-1}(\partial v / \partial v) = A_A^{-1}A_A(v|_{\partial D}),$$

whence $v|_{\partial D} = 0$ by the preceding lemma. This implies $v = 0$ and $u = 0$.

In order to prove existence of a solution we first choose $u_1, v_1 \in H^1(D)$ satisfying $\operatorname{div}(A\nabla u_1) - \gamma u_1 = h_1$ and $A v_1 - v_1 = h_2$ in D and the homogeneous boundary conditions $u_1|_{\partial D} = v_1|_{\partial D} = 0$.

The preceding lemma also ensures that the equation

$$\tilde{f} = A_A^{-1}A_A(\tilde{f}) + A_A^{-1}(\tilde{g}) - f \quad (5)$$

has a unique solution $\tilde{f} \in H^{1/2}(\partial D)$ which depends continuously on $A_A^{-1}(\tilde{g}) - f$. Here, we have abbreviated $\tilde{g} := g - ((\partial u_1 / \partial v_A) - (\partial v_1 / \partial v))$. Defining v_2 as the solution to $A v_2 - v_2 = 0$ in D with $v_2|_{\partial D} = \tilde{f}$ and u_2 as the solution to $\operatorname{div}(A\nabla u_2) - \gamma u_2 = 0$ in D with $\partial u_2 / \partial v_A = A_A(\tilde{f}) + \tilde{g}$ we compute that $u = u_1 + u_2, v = v_1 + v_2$ satisfy the differential equations. Furthermore, we see

$$u|_{\partial D} - v|_{\partial D} = u_2|_{\partial D} - v_2|_{\partial D} = A_A^{-1}(A_A(\tilde{f}) + \tilde{g}) - \tilde{f} = f$$

due to Eq. (5) and

$$\begin{aligned} \frac{\partial u}{\partial v_A} - \frac{\partial v}{\partial v} &= \frac{\partial u_2}{\partial v_A} - \frac{\partial v_2}{\partial v} + \left[\frac{\partial u_1}{\partial v_A} - \frac{\partial v_1}{\partial v} \right] \\ &= A_A(\tilde{f}) + \tilde{g} - A_A(\tilde{f}) + \left[\frac{\partial u_1}{\partial v_A} - \frac{\partial v_1}{\partial v} \right] \\ &= g. \end{aligned}$$

Hence, u and v also satisfy the boundary conditions. Finally, for the norm inequality we observe the estimate

$$\|u_1\|_{H^1(D)} + \|v_1\|_{H^1(D)} + \|\operatorname{div}(A\nabla u_1)\|_{L^2} + \|A v_1\|_{L^2} \leq \tilde{C}\{\|h_1\|_{L^2} + \|h_2\|_{L^2}\}.$$

Consequently, $\tilde{g}$ depends continuously on the data, and so does $\tilde{f}$. This implies the continuous dependence of u and v on the data. $\square$

Now, we conclude this section with a technical result about the interior transmission problem (ITP) that we need during the analysis of the inverse problem.

Lemma 5. Suppose A satisfies assumption (4) and $u_k, v_k, k \in \mathbb{N}$, denote a sequence of solutions to the interior transmission problem (ITP) with boundary data f_k and g_k . If the sequences (f_k) and (g_k) converge in $H^{1/2}(\partial D)$ and $H^{-1/2}(\partial D)$, respectively, and if the sequences $(u_k), (v_k)$ are bounded in $H^1(D)$, then there exists a subsequence $(v_{k(l)})$ of (v_k) which converges in $H^1(D)$.

Proof. Due to the compact imbedding of $H^1(D)$ into $L^2(D)$ we can select L^2 -convergent subsequences $(u_{k(l)})$ and $(v_{k(l)})$. From the differential equations

$$\operatorname{div}(A \nabla u_{k(l)}) - \gamma u_{k(l)} = -(\gamma + \kappa^2 n) u_{k(l)}, \quad \Delta v_{k(l)} - v_{k(l)} = -(\kappa^2 + 1) v_{k(l)},$$

and the boundary conditions

$$u_{k(l)}|_{\partial D} - v_{k(l)}|_{\partial D} = f_{k(l)}, \quad \frac{\partial u_{k(l)}}{\partial \nu_A} - \frac{\partial v_{k(l)}}{\partial \nu} = g_{k(l)},$$

together with the well-posedness of the problem (MITP) we infer the assertion from the last theorem. $\square$

4. The inverse problem

Let us consider the following inverse problem related to the transmission problem (TP): given, on a large sphere ∂B surrounding $\bar{D}$, the scattered solutions $w|_{\partial B}$ for all incident plane waves $v(x) := e^{ikd \cdot x}$, $x \in \mathbb{R}^3$, $d \in S^2 := \{x \in \mathbb{R}^3 : |x| = 1\}$, find the obstacle D . By a solution of (TP) for an incident wave v we mean the solution to (TP) with boundary data $f := v|_{\partial D}$ and $g := \partial v / \partial \nu$. One might be tempted to ask for the reconstruction of the matrix A from these data. However, there exist examples of different matrices A leading to the same data (see [8, Proposition 4.1]), i.e., generally A is not uniquely determined by the Dirichlet data $w|_{\partial B}$. Therefore, we settle for the shape D of the obstacle. The main result of this section states that these data actually suffice to determine D uniquely.

Two of the main ideas to prove this result can already be found in [11,13] where the authors treat the isotropic case. Assuming coincidence of the scattered waves on ∂B for incident plane waves, one first establishes this coincidence for incident waves having a singularity. Next, assuming two different domains D and $\tilde{D}$ one derives a contradiction by using these incident waves with a singularity and moving the singularity closer and closer to the boundary of one domain while the singularity stays away from the other domain. The analysis in [13,3] uses boundary integral equation techniques and therefore it has only been applicable to problems where the leading coefficients of the partial differential operators are constant (at least at the boundary). Our proof employs Lemma 5 about the interior transmission problem instead, thus avoiding the boundary integral equations. The use of the interior transmission problem also distinguishes Isakov's proofs in [10,11] from our proof.

Theorem 6. *Let the domains D and $\tilde{D}$, the matrix-valued functions A and $\tilde{A}$, and the functions n and $\tilde{n}$ satisfy the assumptions of (TP) and Assumption (4). If for all incident plane waves the scattered solutions $w|_{\partial B}$, corresponding to the data D , A , n , and $\tilde{w}|_{\partial B}$, corresponding to $\tilde{D}$, $\tilde{A}$, $\tilde{n}$, coincide on a large sphere ∂B , i.e., $w|_{\partial B} = \tilde{w}|_{\partial B}$, then we have $D = \tilde{D}$.*

Proof. As in (3) we denote by $\Phi(x, y)$ the fundamental solution to the Helmholtz equation, and by Z we denote the unbounded, connected component of $\mathbb{R}^3 \setminus (\bar{D} \cup \bar{\tilde{D}})$. In a first step we show that the scattered solutions $w|_{\partial B}$ and $\tilde{w}|_{\partial B}$ also coincide for the incident waves $\Phi(\cdot, z)$ with $z \in Z$ being a parameter.

For points z from the exterior of the closed ball $\bar{B}$ the function $\Phi(\cdot, z)$ is a solution to the Helmholtz equation in a neighborhood of $\bar{B}$. Hence, we can approximate $\Phi(\cdot, z)$ by plane incident

waves with respect to the $H^1(B)$ -norm (see [13, Lemma 3.2]). Then, the well-posedness of (TP) implies the coincidence of $w|_{\partial B}$ and $\tilde{w}|_{\partial B}$ for these incident waves. Since the functions $\Phi(\cdot, z)|_{\partial D}$ and $\partial\Phi(\cdot, z)/\partial v|_{\partial D}$ depend in a real-analytic way on the parameter z , we can again conclude from the well-posedness of (TP) that the boundary values $w|_{\partial B}$ and $\tilde{w}|_{\partial B}$ depend in a real-analytic manner on z , and therefore, that they must coincide for all $z \in Z$.

Let us now assume that $\bar{D} \not\subset \tilde{D}$. Since $\tilde{D}_e$ is connected, we can find a point $z \in \partial D$ and $\varepsilon > 0$ with the following properties:

- (I) $B_{8\varepsilon}(z) \cap \tilde{D} = \emptyset$,
- (II) the intersection $\bar{D} \cap B_{8\varepsilon}(z)$ is contained in the connected component of $\bar{D}$ to which z belongs,
- (III) there are points from this connected component of $\bar{D}$ to which z belongs, which are not contained in $\bar{D} \cap \tilde{B}_{8\varepsilon}(z)$,
- (IV) the points $z_k := z + (\varepsilon/k)v(z)$ lie in Z for all $k \in \mathbb{N}$.

Due to the singularity of $\Phi(\cdot, z_k)$ at the point z_k we have $\|\Phi(\cdot, z_k)\|_{H^1(D)} \rightarrow \infty$, $k \rightarrow \infty$. In the sequel we will consider the incident waves

$$v_k(x) := \frac{1}{\|\Phi(\cdot, z_k)\|_{H^1(D)}} \Phi(x, z_k), \quad x \in \bar{D} \cup \tilde{D},$$

and the corresponding solutions u_k , w_k and $\tilde{u}_k$, $\tilde{w}_k$. It is our aim to prove that the above assumption, $\bar{D} \not\subset \tilde{D}$, and the equality $w|_{\partial B} = \tilde{w}|_{\partial B}$ allows the selection of a subsequence $(v_{k_j})_j$ from $(v_k)_k$, which converges to zero with respect to $H^1(D)$. This certainly contradicts the definition of the v_k .

Since the functions $\Phi(\cdot, z_k)$ together with their first and second derivatives are uniformly bounded in every compact subset of $\mathbb{R}^3 \setminus B_{2\varepsilon}(z)$, we know $\|v_k\|_{H^1(\tilde{D})} \rightarrow 0$, $k \rightarrow \infty$, whence $\tilde{w}_k|_{B \cap Z} \rightarrow 0$, $k \rightarrow \infty$, by the well-posedness of (TP) for the domain $\tilde{D}$. From $w_k|_{\partial B} = \tilde{w}_k|_{\partial B}$, the uniqueness of the exterior Dirichlet problem for the Helmholtz equation in $\mathbb{R}^3 \setminus B$, and the analyticity of solutions to the Helmholtz equation we conclude $w_k|_{B \cap Z} = \tilde{w}_k|_{B \cap Z} \rightarrow 0$, $k \rightarrow \infty$, with respect to $H^1(Z \cap B)$. With the help of a cutoff function $\chi \in C_0^\infty(B_{8\varepsilon}(z))$ satisfying $\chi(x) = 1$ in $B_{7\varepsilon}(z)$ this means

$$(\chi w_k)|_{\partial D} \rightarrow 0, \quad \frac{\partial(\chi w_k)}{\partial v} \rightarrow 0, \quad k \rightarrow \infty, \quad (6)$$

with respect to the $H^{1/2}$ -norm and the $H^{-1/2}$ -norm on ∂D , respectively. For the convergence of $\partial(\chi w_k)/\partial v$, first, we must derive the convergence of $\Delta(\chi w_k)$, $k \in \mathbb{N}$, in $L^2(B \cap D_e)$ which follows from $\Delta(\chi w_k) = \chi \Delta w_k + 2\nabla \chi \cdot \nabla w_k + w_k \Delta \chi$, $\Delta w_k = -\kappa^2 w_k$, and the H^1 -convergence of the sequence (w_k) in $B \cap Z$. Note, that here we need conditions (II) and (IV) on z in order to ensure $B_{8\varepsilon}(z) \cap D_e = B_{8\varepsilon}(z) \cap Z$.

Next, in the exterior of $B_{2\varepsilon}(z)$ the $H^2(B \setminus B_{2\varepsilon}(z))$ -norms of v_k remain uniformly bounded. Then, the assertion about the boundary regularity of the scattered wave made in Theorem 2(b) implies $w_k|_{(B \cap D_e) \setminus B_{4\varepsilon}(z)} \in H^2((B \cap D_e) \setminus B_{4\varepsilon}(z))$ has uniformly bounded H^2 -norms. Therefore, using the compact imbedding of $H^2(B \cap D_e)$ into $H^1(B \cap D_e)$, we can select a $H^1(B \cap D_e)$ convergent subsequence $((1 - \chi)w_{k_j})_j$ from $((1 - \chi)w_k)_k$. Hence, $((1 - \chi)w_{k_j})|_{\partial D}$, $j \in \mathbb{N}$, is a convergent sequence in $H^{1/2}(\partial D)$, and similarly to the above reasoning we also have convergence of $\partial((1 - \chi)w_{k_j})/\partial v$, $j \in \mathbb{N}$, in $H^{-1/2}(\partial D)$.

Together with the convergence from (6) we conclude that the sequences

$$(w_{k_j}|_{\partial D})_j, \left(\frac{\partial w_{k_j}}{\partial \nu} \right)_j,$$

converge in $H^{1/2}(\partial D)$ and $H^{-1/2}(\partial D)$, respectively.

Since the functions u_k and v_k are a solution to the interior transmission problem (ITP) for the domain D with boundary data $w_k|_{\partial D}$ and $\partial w_k/\partial \nu$, and since the $H^1(D)$ -norms of u_k and v_k remain uniformly bounded, according to Lemma 5 we can select a subsequence of the $(v_{k_j})_j$ which converges in $H^1(D)$. Let us again denote by $(v_{k_j})_j$ this convergent subsequence and by v_0 its limit. v_0 is a weak solution to the Helmholtz equation in D because it is the limit of weak solutions to the Helmholtz equation. Furthermore, we have $v_0|_{D \setminus B_{2\epsilon}(z)} = 0$ because the functions v_k uniformly converge to zero in the exterior of $B_{2\epsilon}(z)$. Then, v_0 must be zero in all of D (here we make use of condition (III), namely the fact that the connected component of D containing z has points which do not lie in the exterior of $\overline{B_{2\epsilon}(z)}$). This contradicts the fact $\|v_k\|_{H^1(D)} = 1$, $k \in \mathbb{N}$, from the definition of the v_k . Hence, our assumption $\bar{D} \not\subset \tilde{\bar{D}}$ is false.

Since we can derive the analogous contradiction for the assumption $\tilde{\bar{D}} \not\subset \bar{D}$, we have proved $D = \tilde{D}$. $\square$

Remark. Since it is possible to compute the far field pattern of w from $w|_{\partial B}$ and *vice versa* the last theorem also shows that D is uniquely determined by the far fields of the scattered waves for all incident plane waves.

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